

Build an IR Remote-Controlled Lamp with a CD4017

Introduction

Have you ever turned off the light in your room and then realized that you still need to get out of bed to turn it back on? Sometimes, the simplest things can become surprisingly inconvenient.

What if you could control a **220 V lamp from a distance using a simple infrared remote control**, without using a microcontroller?

In this project, we will build a simple **IR remote-controlled lamp** using a **TSOP38 infrared receiver**, a **CD4017 decade counter**, and a **relay**.

The system works with a simple push of a button on the remote: each press changes the state of the lamp. The same IR emitter developed in the previous Instructable can be used as the transmitter, making this project the next step in our IR system.

Beyond convenience, this approach can also be useful wherever we want to control an electrical load **without having to physically operate its switch**.

The circuit is intentionally built without a microcontroller, using simple and accessible electronic components to demonstrate how an infrared signal can be transformed into a practical control system.

What You'll Learn

By the end of this tutorial, you will be able to:

- Understand how an **IR receiver and a CD4017** can be used to control a lamp remotely.
- Configure a **CD4017 as a toggle controller** for a single lamp.
- Control a **220 V lamp through a relay**.

Supplies

- 1 × CD4017
- 1 × TSOP38 IR receiver
- 1 × LED
- 1 × Electromagnetic relay
- 2 × 1N4007 diode
- 1 × 2N2222 transistor
- 1 × 100 nF capacitor
- 2 × 100 uF capacitor
- 2 × 220 uF capacitor

- 2× 10 kΩ resistor
- 1 × 20 kΩ potentiometer
- 1 × LM358
- 1 × 78L05
- 1 × Breadboard/ Prototyping Veroboard
- Jumper wires for the breadbord

Step 1: Understand the Circuit

The system is built around five main blocks:

IR Receiver → Comparator → CD4017 → Relay Driver → Relay → Lamp

The **IR receiver** detects the 38 kHz infrared signal from the remote control and produces an electrical pulse at its output.

The **LM358 comparator** conditions this signal by comparing the receiver's output with a reference voltage, producing a clean logic-level pulse suitable for the CD4017.

The **CD4017** acts as a decade counter and changes its output state each time it receives a clock pulse. This allows each press of the remote control to toggle the lamp between ON and OFF.

The **relay driver**, built around a **2N2222 transistor**, provides the current required to energize the relay coil.

Finally, the **relay** switches the 220 V supply to the lamp, allowing the low-voltage control circuit to operate the AC load safely.

The block diagram below shows the complete system:

IR Remote



IR Receiver



LM358 Comparator



CD4017



2N2222 Relay Driver



Relay



220 V Lamp

Step 2: Configure the CD4017

The **CD4017** is a decade counter. Each valid clock pulse advances the HIGH state from one output to the next:

Q0 → Q1 → Q2 → Q3 → ... → Q9 → Q0

For the counter to advance:

- **Clock Inhibit = LOW** → clock pulses are accepted.
- **Clock = rising edge** → the counter advances on each rising edge.
- **Master Reset = LOW** → normal counting operation.

The **Master Reset** input is used to force the counter back to **Q0**. When it is HIGH, the counter immediately returns to Q0, regardless of the clock.

The **Clock Inhibit** input controls whether the counter accepts clock pulses. When HIGH, the counter stops counting.

In our circuit, we will use **Q0, Q1, and Q2** to control the lamp and create the desired ON/OFF sequence.

Step 3: Create the Lamp Control Sequence

We only use **Q0, Q1, and Q2** of the CD4017 to create the ON/OFF sequence.

Clock Pulse	Active Output	Lamp
Initial state	Q0	ON
1st press	Q1	OFF
2nd press	Q2	OFF
Reset	Q0	ON

When **Q2 becomes HIGH**, it activates the **Master Reset**, forcing the CD4017 back to Q0.

This creates a simple **toggle function**: each press of the remote changes the state of the lamp.

Step 3: Condition the IR Receiver Signal

The TSOP38 output changes state when it detects the **38 kHz IR signal**.

However, the receiver produces a rapidly changing signal during the IR burst. We only want the CD4017 to receive **one clock pulse per button press**, not multiple pulses.

To achieve this, we use a simple **pulse-conditioning circuit**:

- A **100 nF capacitor** stores the pulse.
- A **10 kΩ resistor** provides a discharge path to ground.
- A **diode** isolates the capacitor from the TSOP38 output and prevents the receiver from discharging the capacitor during the 38 kHz bursts.

This converts the fast IR receiver activity into a single pulse suitable for the CD4017 clock input.

Step 5: Build the Circuit

Now that we understand how each part of the system works, we can build the complete circuit.

Use the schematic below as a guide to connect:

- The **TSOP38 IR receiver**
- The **pulse-conditioning circuit**
- The **CD4017**
- The **2N2222 relay driver**
- The **1N4007 protection diode**
- The **relay**
- The **LED indicator**

The complete circuit and the finished prototype are shown below.

Step 6: Test the Circuit

Before testing the remote control, first adjust the **detection threshold** of the LM358 comparator.

Use the **resistive trimmer** connected to the **non-inverting input (+)** of the comparator to set the reference voltage to approximately **3 V**. Measure the voltage with a multimeter and adjust the trimmer until the desired threshold is reached.

Once the threshold is correctly adjusted, point the IR transmitter toward the TSOP38 receiver and press the remote-control button.

Each press should toggle the lamp:

- **1st press → Lamp ON**
- **2nd press → Lamp OFF**

Verify that the circuit responds reliably to **one press at a time** and that the relay switches correctly.

 **Safety:** The relay contacts are connected to **220 V mains voltage**. Disconnect the power before making any changes to the circuit, and use proper insulation and electrical safety practices.